

MD. MOSFIKUR RAHMAN

Senior Backend Engineer | .NET & Java

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SUMMARY

Senior Backend Engineer with 4+ years of production experience across .NET and Java (Spring Boot) ecosystems, currently owning end-to-end backend architecture for a multi-tenant OTA booking SaaS platform spanning 7 .NET 9 microservices – taken from an empty repository to two live white-label client deployments in 13 months. Lead a team of 10 engineers as principal author of the platform backend. Design distributed, service-oriented systems – REST, GraphQL, gRPC, and SSE APIs – and provide technical direction and mentorship across backend, frontend, and DevOps. Prior experience spans government compliance systems (Java, Spring Boot, Oracle) and e-commerce platforms (GraphQL BFF for Rakuten), with production deployments on Docker, Kubernetes, Azure, and AWS. Grounded in a research background of 10 peer-reviewed publications in applied machine learning.

EXPERIENCE

Akij iBOS Ltd.

Software Engineer L-II (Team Lead)

Dhaka, Bangladesh

Nov. 2024 – Present

- Backend architect for Travilo, a multi-tenant OTA booking SaaS platform, owning end-to-end design across 7 .NET 9 microservices for supplier configuration, pricing/markup/discount rules, ancillary services, and booking workflows; principal author of the platform backend and top contributor on six of the seven services.
- Lead a team of 10 engineers across backend, frontend, and DevOps – personally training and mentoring junior engineers, running daily scrum/sprint planning, and conducting code reviews and pair programming that established standardized backend patterns; the second white-label client went live six weeks after the first.
- Own the admin control plane – 58 modules across 167 routes, with a role-based access model spanning 175 menus and 55 roles – giving operations teams tenant-level control of pricing, ancillary services, finance, and booking workflows.
- Built a cross-service booking expiration/cancellation scheduler (gRPC + REST across Reservation, Configuration, and Notification services) that auto-expires unticketed bookings and keeps partner-facing state in sync.
- Designed the flight-search aggregation layer over 11 supplier and GDS connectors (Sabre, Amadeus, Travelport, and airline-direct APIs): Reservation fans out over gRPC to AirMaster, which aggregates downstream responses, and streams progressive results to the frontend via Server-Sent Events (SSE).
- Built the platform's observability layer – middleware capturing every incoming and inter-service call (method, path, status, duration, payloads) to blob storage behind a unified Azure/AWS/Cloudflare storage abstraction – now holding 36,000+ structured log objects in production and used daily for production debugging and on-call triage.
- Deploy with Docker for local development and Kubernetes for client-facing cloud environments; use Azure for CI/CD, Kubernetes-pod deployment, and storage, and AWS for transactional email; provide on-call production support, including weekend/emergency response, as booking volume grows past 1,000 per month.

REVE Systems Ltd.

Software Engineer

Dhaka, Bangladesh

Jul. 2023 – Oct. 2024

- Delivered the Legal Case and Utilization Declaration modules for the Customs Bond Management System, a National Board of Revenue (NBR) platform that digitized customs bond compliance for garment exporters, replacing manual processing.
- Collaborated with senior engineers and architects on overall system design while owning implementation of assigned modules end-to-end in Java, Spring Boot, and Oracle.
- Built the BGMEA/BKMEA database sync integration for export compliance, working closely with the Business Analysis team to translate compliance requirements into backend logic.
- Implemented OAuth2 / Spring Security authentication and clean-architecture REST APIs, and improved system throughput by ~25% through query tuning and data-access optimization.

BJIT Group

Software Engineer (Backend, Java & GraphQL)

Dhaka, Bangladesh

Apr. 2022 – Jun. 2023

- Implemented and enhanced a GraphQL Backend-for-Frontend for Rakuten e-commerce modules using Java and Spring Boot, providing a unified API layer over multiple backend services that reduced frontend integration complexity.
- Collaborated with Rakuten engineers and the internal BJIT team on schema design, contributing to technical discussions on resolver optimization and backend implementation approaches.
- Implemented resolver batching and schema stitching, reducing nested-query latency, and optimized database access paths, contributing reusable backend service patterns adopted across the team.

SELECTED PROJECTS

Travilo (OTA Booking Platform)

Akij iBOS Ltd.

Stack: .NET 9, SQL Server, Microservices, DynamoDB, gRPC, GraphQL, SSE

- Backend architect for a live travel platform spanning B2B, B2C, and API-only/consortium distribution – two white-label client deployments in production serving 742 B2B agency accounts across 7 .NET 9 microservices.
- Exposed REST and GraphQL APIs for partner integration, including IP-whitelisted GraphQL access for API-only clients, plus an admin panel for centralized platform management; pricing, markup, discount, and ancillary logic is modelled as tenant configuration rather than code – 96 rules live – so commercial changes ship without a deployment.

TrackForce (Employee Activity Monitoring)

Akij iBOS Ltd.

Stack: ASP.NET Core, Time-Series Storage

- Architected TrackForce, a real-time employee-activity monitoring platform for 500+ employees, designing the end-to-end data flow from the on-device agent through ingestion, storage, and the reporting dashboard.
- Defined the API and service boundaries between the ingestion pipeline and the storage/reporting layers so implementation could proceed against a fixed contract.

Customs Bond Management System (NBR)

REVE Systems Ltd.

Stack: Java, Spring Boot, Thymeleaf, Oracle

- National customs-compliance platform digitizing bond management for garment exporters; delivered the Legal Case and Utilization Declaration modules end-to-end within an architecture owned by senior engineers.
- Built the BGMEA/BKMEA compliance sync and implemented OAuth2/Spring Security authentication with audit-trail logging and role-based access control.

Rakuten Echiba Backend

BJIT Group

Stack: Java, Spring Boot, GraphQL

- GraphQL BFF unifying multiple backend services into a single API layer for Rakuten e-commerce modules, reducing frontend integration complexity.
- Reduced nested-query latency via resolver batching and schema stitching, and optimized database access paths and join patterns.

TECHNICAL SKILLS

Languages	Java, C#, TypeScript/JavaScript, SQL, Python
Java Backend	Spring Boot, Spring Security, Spring Data JPA/Hibernate, Spring GraphQL, JUnit, Mockito
.NET Backend	ASP.NET Core (.NET 9), Entity Framework Core, HotChocolate GraphQL
Architecture	Microservices, Distributed Systems, System Design, Clean/Layered Architecture, Domain-Driven Design, REST APIs, gRPC, SSE
Cloud & DevOps	AWS, Azure, Docker, Kubernetes, CI/CD (GitHub Actions), Azure Monitor/App Insights, ELK/Seq
Databases	PostgreSQL, SQL Server, MySQL, Oracle, MongoDB, DynamoDB
Frontend	Next.js, React, AngularJS, Ant Design, Tailwind
Domains	Online Travel Agency (OTA) & Booking Systems, GDS/Supplier Integration, PNR & Ticketing Workflows, SaaS Platforms, Multi-tenant Systems, E-commerce, Government Compliance Systems
Concepts	OAuth2/JWT, Design Patterns, RBAC & Secure APIs, Query & Performance Optimization, Agile/Scrum, Pair Programming
Tools	Git, Linux/Unix

EDUCATION

Daffodil International University

B.Sc. in Computer Science and Engineering, CGPA: **3.83/4.00**

Dhaka, Bangladesh

2018 – 2021

Adam Mickiewicz University

Erasmus+ International Exchange (one semester)

Poznań, Poland

2021

RESEARCH & PUBLICATIONS

10 peer-reviewed publications (3 journal, 7 conference) in applied machine learning – health analytics, IoT systems, and security; 2 as first author. Reviewer for *ISA Transactions*, *Journal of King Saud University – Computer and Information Sciences*, and *Natural Language Processing Journal*. Selected work:

- **Rahman, Md. Mosfikur**, Kashem, M. A., Mohiuddin, M., Hossain, M. A., & Moon, N. N. (2020). “Future City of Bangladesh: IoT Based Autonomous Smart Sewerage and Hazard Condition Sharing System.” *Proc. 2020 IEEE Intl. Women in Engineering Conf. on Electrical and Computer Engineering (WIECON-ECE)*, pp. 126–130. **IEEE Best Paper Award** · First author · [10.1109/WIECON-ECE52138.2020.9397950](https://doi.org/10.1109/WIECON-ECE52138.2020.9397950)
- **Rahman, Md. Mosfikur**, Saifuzzaman, M., Ahmed, A., Mahin, M. F., & Shetu, S. F. (2021). “Impact of COVID-19 on mental health: A quantitative analysis of anxiety and depression based on regular life and internet use.” *Current Research in Behavioral Sciences* (Elsevier), 2:100037. First author · Journal · [10.1016/j.crbeha.2021.100037](https://doi.org/10.1016/j.crbeha.2021.100037)
- Shetu, S. F., **Rahman, Md. Mosfikur**, Ahmed, A., Mahin, M. F., Akib, M. A. U., & Saifuzzaman, M. (2021). “Impactful e-learning framework: A new hybrid form of education.” *Current Research in Behavioral Sciences* (Elsevier), 2:100038. Most-cited work · Journal · [10.1016/j.crbeha.2021.100038](https://doi.org/10.1016/j.crbeha.2021.100038)
- Islam, T., **Rahman, Md. Mosfikur**, Jabiullah, M. I., & Saifuzzaman, M. (2023). “Cyber Security Intruder Detection Using Deep Learning Approach.” *Information Systems and Management Science*, Springer International, pp. 518–530. Deep-learning intrusion detection on *CICIDS2017* · [10.1007/978-3-031-13150-9_42](https://doi.org/10.1007/978-3-031-13150-9_42)
- Johora, F. T., Mahfuja, I. B., Rahman, A. N. M. M., **Rahman, Md. Mosfikur**, & Rahman, M. S. (2024). “Machine Learning-Based Prediction of COVID-19: A Robust Approach for Early Diagnosis and Treatment.” *Proc. 5th Intl. Conf. on Trends in Computational and Cognitive Engineering*, Springer Nature Singapore, pp. 205–215. Most recent · [10.1007/978-981-97-1923-5_16](https://doi.org/10.1007/978-981-97-1923-5_16)

Full list: [Publication List](#) · [Google Scholar](#).

EXTRA-CURRICULAR ACTIVITIES

- **Conference Speaker**
Delivered a technical session at the 10th IEEE International Women in Engineering (WIE) Conference on Electrical and Computer Engineering 2024 (IEEE WIECON-ECE 2024).
- **Vice President, Research & Career Wing**
Led mentorship programs, research workshops, and career initiatives at the Computer & Programming Club, DIU (2021).
- **Journal Reviewer**
Reviewer for several Elsevier/Springer journals and international conferences in ML and data science.